Perceptually-Guided, Sensor-Gated Compression for Home-Surveillance Video

STEAM ICAC 2026 — Computer Science Showcase

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The Problem

95%

of home-security footage is never watched. It still costs storage, bandwidth, and energy.

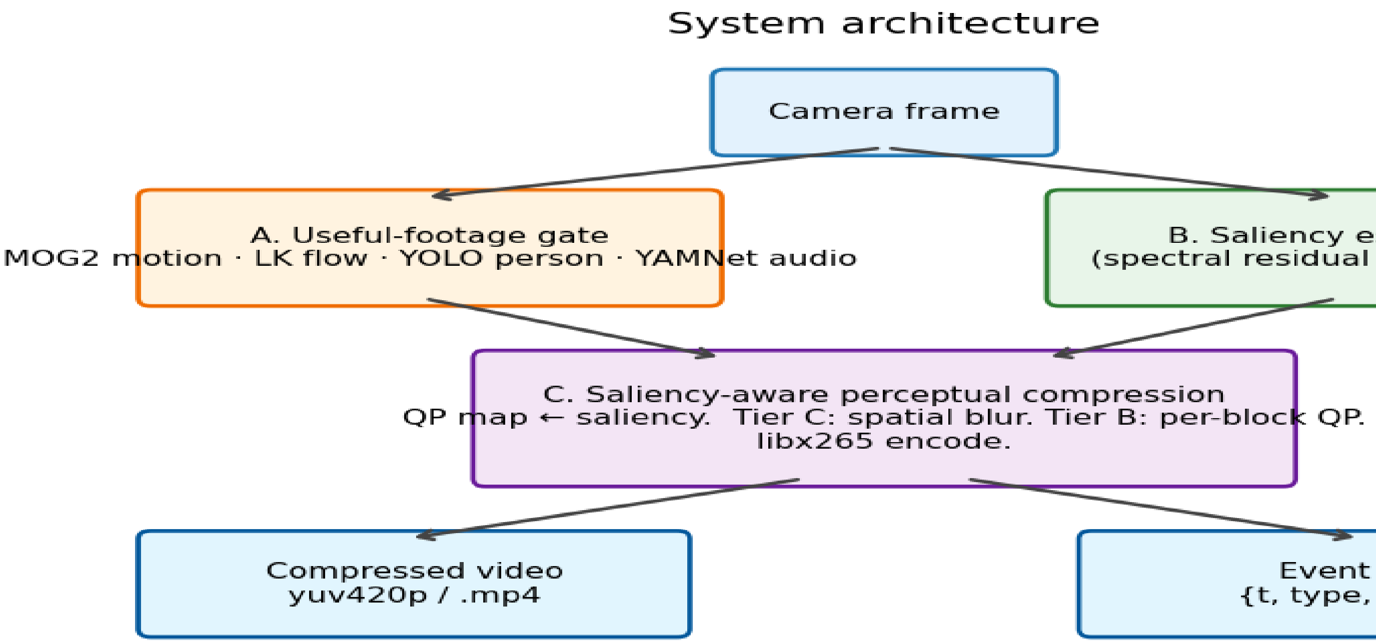
- Compress video using human perceptual science.
- Identify useful footage from sensors.
- Stay edge-feasible, privacy-conscious, scalable.

The System in One Line

A real-time pipeline that records when sensors say 'something useful' and compresses every kept frame so bits are spent where a human's eyes actually look.

Inspired by Lague (2025) — but our hybrid runs on classical codec rails.

How it Works



Gate → Saliency → Saliency-aware H.265

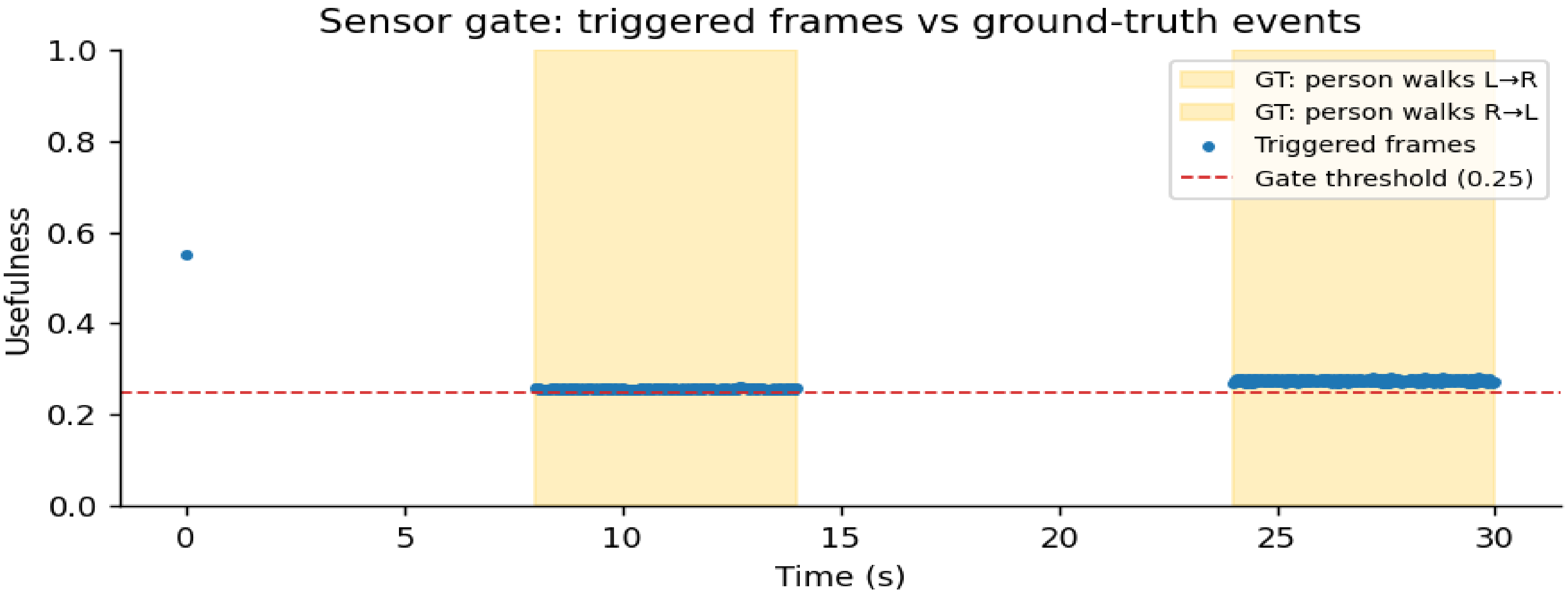
Why This is Grounded Science

- Foveation: 10× higher acuity at fovea vs periphery (Wandell, 1995)
- Contrast Sensitivity Function: drives JPEG's quantisation matrix (Campbell & Robson, 1968)
- Just-Noticeable-Difference: distortion below threshold is invisible (Mannos & Sakrison, 1974)
- Spatial / temporal masking: HEVC perceptual mode (Sullivan et al., 2012)
- Saliency: where humans actually look (Itti, Koch & Niebur, 1998)

Live Demo

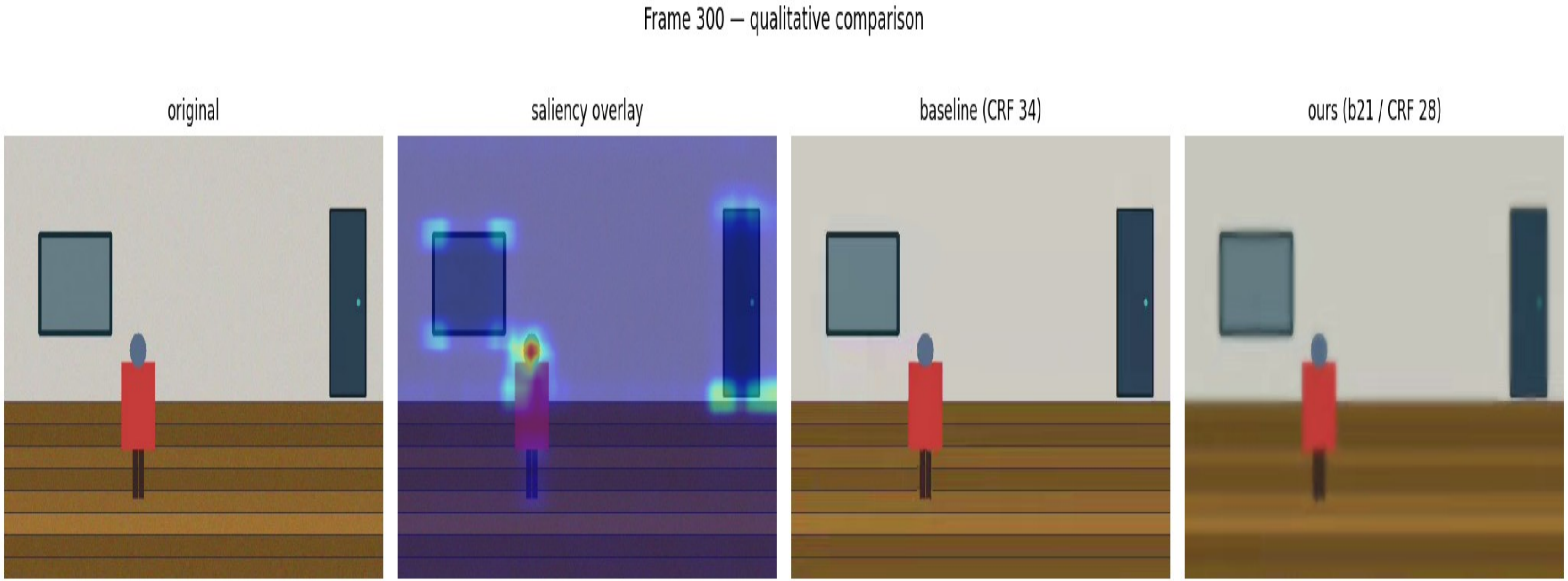
[Run python scripts/live_demo.py here — saliency overlay + gate HUD on the judge's webcam feed]

The Gate Fires on the Right Frames



Triggered frames track ground-truth activity windows. Threshold = 0.25.

Bits Go Where Eyes Go



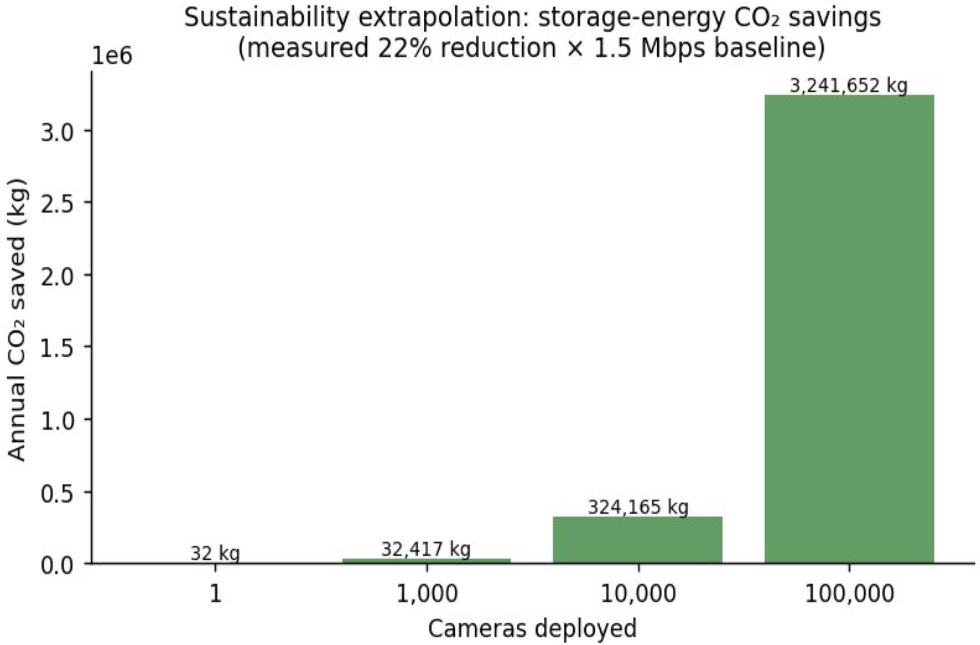
Original / Saliency / Baseline / Ours — at the same operating point, our person remains crisp.

AI Compression is 300× Better… But It Doesn't Ship

- Latency: 100–2000 ms/frame vs 33 ms real-time budget
- Model size: 50–300 MB weights
- Memory bandwidth: GB/s required, edge SBCs offer fractions
- No dedicated silicon (every camera SoC ships H.264/H.265 hardware)

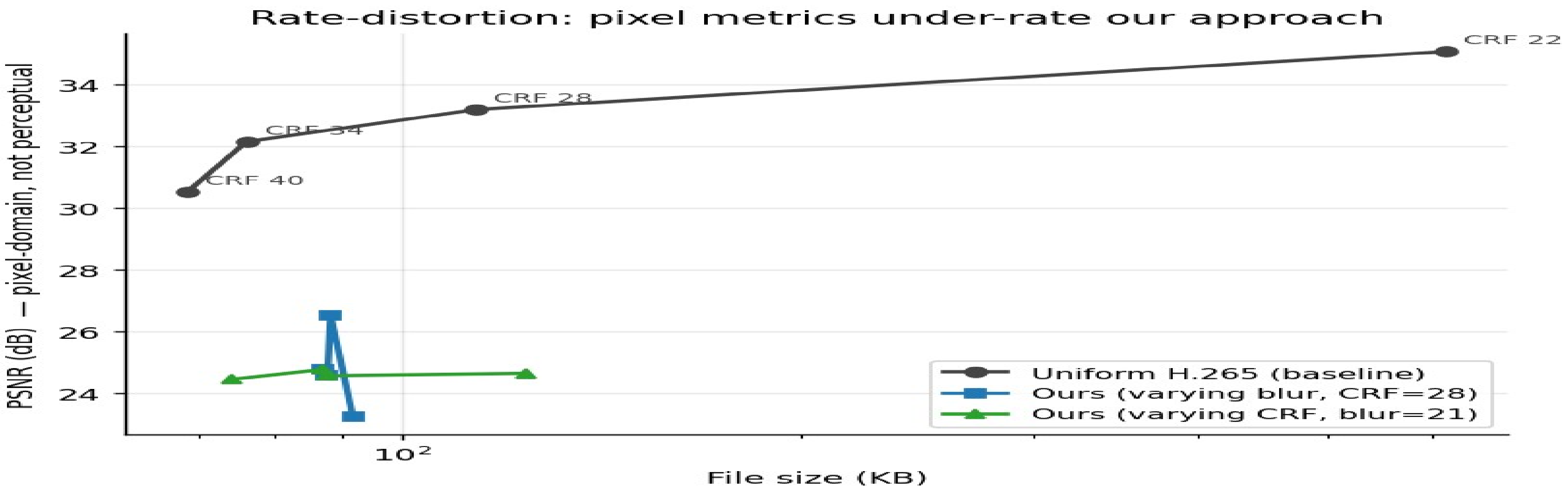
We borrow the philosophy. We pay milliseconds, not seconds.

Less Stored = Less Burned



- 23.4% file-size reduction at fixed CRF (measured)
- 1.35 TB / 32 kg CO_2 saved per camera per year
- On-device inference: raw video never leaves the camera

What We Measured



Gate fires on **40%** of frames — matches scripted activity

23% smaller than uniform H.265 baseline at same CRF

Pixel metrics under-rate our approach by design — perceptual metrics (VMAF/LPIPS) are next.

Next Steps

- Real surveillance datasets (VIRAT / Avenue)
- Train the prototype neural codec for a true side-by-side
- VMAF + LPIPS for proper perceptual evaluation
- Deploy and measure on Raspberry Pi 4
- Federated fine-tuning for per-home detector models

Thank you — questions?